Curriculum Vitae

Benjamin J. Wilson

Personal Details

Address	Institut de Mathématiques de Jussieu, 175 Rue du Chevaleret, Paris, 75013, France
Telephone	+33 6 61 65 23 74 , +61 3 63 44 94 70
E-mail	wilson@math.jussieu.fr
Citizenship	Australian
Date of Birth	January 8, 1982

Academic History

10/2008 – 9/2009	**Post-doctoral scholar** at the Institut de Mathématiques de Jussieu, Université de Paris VII, under the supervision of Éric Vasserot.
2004 – 2008	**Ph.D. (Mathematics)** at the University of Sydney (Australia) and the Universidade de São Paulo (Brazil), under the supervision of Alexander Molev and Vyacheslav Futorny, respectively. Graduated June 6, 2008. Thesis entitled *Representations of Infinite-Dimensional Lie Algebras*.
2003	**Honours in Pure Mathematics** at the University of Sydney, under the supervision of Vyacheslav Futorny. Dissertation entitled *Lie Conformal Algebras*. Graduated with **First Class Honours** and the **University Medal**.
2000 – 2002	**B.Sc.(Advanced)** at the University of Sydney. Graduated with majors in Mathematics and Computer Science.

Research Interests

I am broadly interested in algebraic representation theory and categorification. My area of expertise is the representation theory of infinite-dimensional Lie algebras, focussing, in particular, upon highest-weight representation theory, representations of polynomial Lie algebras, and loop modules.

Academic Honours

University Medal of the University of Sydney, awarded for excellence in Honours level study, 2003.

B.H. Neumann Prize of the Australian Mathematical Society for the most outstanding talk by a postgraduate student at the Annual Meeting, 2006.

University Postgraduate Award for postgraduate studies in Mathematics at the University of Sydney (scholarship/stipend), 2004 – 2007.

Academic Visits

Visitor of Prof. Yufeng Zhao, Peking University, April 2009.

Visiting Scholar at the University of California, Riverside, January and February 2008, under the invitation of Prof. Vyjayanthi Chari.

Publications

All articles are available online on the arXiv: `http://front.math.ucdavis.edu/`

A Character Formula for the Category $\tilde{\mathcal{O}}$ of Modules for Affine sl(2), Int. Math. Res. Not., 2008. Available online as `arXiv:0711.0727`.

Imaginary Highest-Weight Representation Theory and Symmetric Functions, accepted by Comm. Algebra, October 2007. Available online as `arXiv:math/0702855`.

Highest-Weight Theory for Truncated Current Lie Algebras, submitted to J. Algebra, 2007. Available online as `arXiv:0705.1203`.

Representations of Truncated Current Lie Algebras, B.H. Neumann Prize Article, Bull. Aust. Math. Soc., 34(5):279–282, 2007. Available online as `arXiv:0706.2923`.

Representations of Infinite-Dimensional Lie Algebras, Ph.D. thesis, available on request.

Lie Conformal Algebras, Honours dissertation, University of Sydney, available on request.

Talks

Character Formulae for the Category $\tilde{\mathcal{O}}$ of Chari, delivered at:

Algebra seminar of the University of California at Riverside, February 2008;

Séminaire d'algèbre, Institut Henri Poincaré, October 2008.

Highest-Weight Theory for Polynomial Lie Algebras, delivered at:

Annual Meeting of the Australian Mathematical Society, winning the B.H. Neumann Prize, September 2006;

Algebra seminar of the University of Sydney, October 2006;

International conference entitled *Lie and Jordan Algebras, their Representations and Applications III*, Maresias, Brazil, August 2007;

Séminaire sur les Algèbres Enveloppantes et Théorie des Représentations at the Institut de Mathématiques de Jussieu, Paris, November 2008.

Imaginary Highest-Weight Representation Theory, delivered at:

Colóquio IME-USP, Universidade de São Paulo, July 2005 (in Portuguese);

International conference entitled *Infinite-dimensional Lie algebras and their Applications*, at the Harish-Chandra Research Institute, Allahabad, India, December 2005;

Algebra seminar of the University of Sydney, March 2006.

Cayley Automata, delivered to the Sydney University Mathematical Society, June 2006.

Teaching

Semester 1, 2008
Lecturer (full-time) in the School of Mathematics and Statistics at the University of Sydney.

Semester 1, 2007
Lecturer (regular casual) of the first-year course *Differential Calculus* at the University of Sydney, with a responsibility for 502 students.

Summer 2007/2008
Instructor of the preparatory mathematics course for entering undergraduates at the University of Sydney, organised by the School of Mathematics and Statistics.

Semesters 1 and 2, 2006
Tutor for the School of Mathematics and Statistics at the University of Sydney. Instructed the first-year courses *Discrete Mathematics* and *Linear Algebra*.

Additional Information

Member of the Australian Mathematical Society.

Fluent speaker of Portuguese; intermediate French.

Proficient in the computer algebra package *Magma* and in the programming languages *Java*, *C++*, *Prolog*.

Referees

Research
Prof. Vyacheslav Futorny
Instituto de Matemática e Estatística
Universidade de São Paulo
futorny@ime.usp.br
+55 11 3091 6294

Research
Prof. Yuly Billig
School of Mathematics and Statistics
Carleton University
billig@math.carleton.ca
+1 613 520-2600 Ext. 4310

Teaching and Employment
Ms. Jenny Henderson
School of Mathematics and Statistics
University of Sydney
J.Henderson@maths.usyd.edu.au
+61 2 9351 2005

Research
A/Prof. Alexander Molev
School of Mathematics and Statistics
University of Sydney
A.Molev@maths.usyd.edu.au
+61 2 9351 5793